

**3.1. Framework and parameters setting.** In this part, we consider the parameters appearing in eqs. (1.5) and (1.6) and define the ranges where they vary. In doing that, we take advantage of the estimates provided in [36] for *prostate cancer treated with androgen deprivation therapy (ADT)*. We recall that the analysis in [36] relied on the results of the trial described in [11]. There, a cohort of patients with biochemical recurrence of the tumor (non-metastatic and castration sensitive, m0CSPC) after radical radiotherapy treatment underwent intermittent ADT.

*Parameters not affected by uncertainty.* We begin by listing the parameters that in our experiments are assumed to be unaffected by uncertainty. The first crucial quantity to be estimated is the proliferation rate for sensitive cells  $r_S$ , which is employed to perform the time-normalization of the original system (1.2) into (1.5). In [36], the authors set  $r_S = 0.027 \text{ day}^{-1}$ , adopted from a previous estimate derived in [40]. Then, the effectiveness of the therapy is encoded in the value of the non-dimensional constant  $\hat{d}_D$ , which is set  $\hat{d}_D = 1.5$  in [36] (see also [38] for the original estimate). We recall that the normalized dimension of the tumor is expressed using  $n = n(\tau)$ , which ranges in  $[0, 1]$ . Here,  $n = 0$  means that the tumor cells (sensitive and resistant) are entirely extinct, while  $n = 1$  implies that the tumor size has reached the system's carrying capacity. The initial size of the tumor  $n(0) = n_0$  in our experiments takes values in  $\{0.25, 0.50, 0.75\} \ni n_0$ . We insist that  $n_0$  is not affected by uncertainty, and we shall run the simulations separately for the different values of  $n_0$ . Finally, in [36] the turnover rate  $\hat{d}_T$  is proposed to range in the interval  $\hat{d}_T \in [0, 0.5]$ . In the present paper, we consider the most adverse scenario, i.e., we always set  $\hat{d}_T = 0$ .

*Parameters affected by uncertainty.* We now describe the parameters that are affected by uncertainty. The first quantity is the normalized proliferation rate of the resistant tumor cells encoded in  $\hat{r}_R$ . In [36] this constant is suggested to vary in the interval  $[0.5, 1] \ni \hat{r}_R$ . Here,  $\hat{r}_R = 0.5$  means that the 'cost' of having the drug resistance results in the proliferation rate being 50% smaller than that of sensitive cells. Conversely, if  $\hat{r}_R = 1$ , the resistant cells have no disadvantage towards the sensitive ones regarding reproduction rate.

In the ODE model considered here, there is no mechanism of resistance acquirement during the evolution. In other words, in our framework, the population of resistant cells  $r(\tau)$  at an instant  $\tau > 0$  consists of clones of an initial resistant subpopulation that is assumed to be present from the very beginning. This hypothesis is consistent with the heterogeneity and polyclonal nature of solid cancers, which in the clinical practice is particularly realistic in the case of advanced or metastatic disease (see e.g. [37]). We assume that the initial fraction  $\hat{f}_0$  of resistant cells ranges in  $[0.002, 0.1] \ni \hat{f}_0$ , meaning that the ratio  $\hat{f}_0 := r(0)/(r(0) + s(0))$  varies between 0.2% (most favourable scenario) and 10% (worst-case scenario). In future work, we plan to incorporate a feature that accounts for the secondary resistance mechanism into the model.

*Space of parameters and ensemble measure.* We are in a position to provide a formal definition for the space of parameters  $\Theta$  where  $\theta := (\hat{d}_D, \hat{d}_T, \hat{r}_R, \hat{f}_0)$  takes values. According